

Database Management Systems

CPS 3740

Lecture 17

Computer Science, Kean University

Dr. Ching-yu (Austin) Huang

Instructor Paolien Wang

Outline

- Lab and Project due
- Last withdraw date soon
- Exam Policy
- Quick Midterm2 Review
- Similar to midterm 1
 - 75 minutes

Exam policy

- **Exam Format:** The exam will be conducted **in person** in the classroom using the **LockDown Browser**.
- **Preparation:** Bring your laptop and charger. Ensure your laptop is updated and **test the LockDown Browser using the practice item before the exam.**
- **Punctuality:** **Arrive at least 5 minutes early.** Attendance will be taken before the exam begins.
- **Prohibited Items:** Headphones, mobile phones, and other electronic devices are not allowed during the exam. **Mobile phones may only be used for Canvas login verification** before the exam starts and must then be stored away.
- **Room Policy:** **Once the exam begins, students cannot leave the room and return to continue their exam.** Leaving the exam room will indicate that you have completed your exam.

Review midterm 2 coverage

- Midterm 2 also covers the material from Midterm 1.
- SQL
 - SELECT, group by, having, natural join, outer join
 - Set, subquery
 - Insert, update, delete with foreign keys
 - Create tables, keys, drop, alter
 - Relational model and algebra
- Project & homework
- Team research
 - GUI, SQL, functions

Advisement & early registration

- Prerequisite for the new Options
 - **Cybersecurity**
 - CPS 3498 Computer Security (fall/spring), Digital Forensics (fall)
 - **Data Science**
 - Data Visualization (fall), Data mining (spring)
 - Math 3700, Math 3526 (+ Math 2416 for Math minor)
 - Both required CPS 3740 Database
 - Contact your CS faculty advisor soon
- 4+1 (5-year programs, BS CS/MS CIS, BS IT/MS CIS)
 - Contact Dr. Liou, CIS graduate coordinator
- CPS/TECH 3740 summer course
- Early registration starts soon
 - Register as soon as it is open depended on your priority.
 - Course filled up quickly and **NO overloaded!**

Midterm 2 Coverage

- True/false, multiple choices
- Coverage: slides day 1 – 16
- Project – HTML, CGI, PHP MySQL
 - SQL injections
 - insert, update, delete in PHP MySQL functions
 - Operations with foreign keys
- Homework
- SQL Statements
- Relational algebra to SQL

SQL Topics – part 1

- SQL DCL, DDL, DML
- CREATE, Keys, constraints
 - PK, unique, FK, cascade, not null
- ALTER, DROP
 - Change table structures
- Inner and outer joins
- Projects, PHP MySQL functions and error handling
 - SELECT, INSERT, UPDATE, DELETE

SQL topics – part 2

- Aggregations
- Subquery
- Joins – inner, outer, natural
- Sets – intersect, except, union
- DML – select, insert, update, delete
- Constraints - foreign key, primary key, not null
- View, Updatable and non-updatable View

Exercises - keys

- Identify the Primary Key
- Create tables with
 - Primary key
 - Foreign key
- Do the dreamhome SQL exercise questions (1 - 116)
 - SELECT, INSERT, UPDATE, DELETE, CREATE, DROP
 - NULL operations
 - NULL and aggregations
 - NOT including REVOKE, GRANT

SQL Exercise – aggregation and subquery

- Use database dreamhome
- All questions should be written in one SQL query
 1. List the number of female and male staff
 2. List the staff name and salary who has the max salary
 3. List the unique city name that branch has female staff
 4. List the street name that has “9” in the postcode
 5. List the number of staff working at London
 6. List staff average salary for each branch

SQL exercises – aggregation, subquery & group by

- Find the staff name who has the maximum salary for each branch
- Find the staff name who has the maximum salary for each city
- Find the branch that does not have any staff
- Find the city that does not have any staff

Subquery exercises – inner join, natural join, outer join

- List all the branch branchno, city, address, and the staff who work in the branch. If the branch has no staff, it should be still in the output.
- List all the course information and the students who took the courses. If the course has no student, it should be still in the output.

Exercises – PHP, HTML, PHP MySQL

- Design a login web page – HTML
- Write the PHP code to catch and display the login
- SQL injection
- INSERT, UPDATE, DELETE in PHP MySQL
 - The return value when the SQL query is successful or failed

Exercises

Select

- Find the number of staff for each branchno
- Find the number of staff for each city
- Find the staff name who has the minimum salary for each branchno
- Find the staff name who has the minimum salary for each city

Exercises

Create, insert, alter, update, delete

- Create a table
 - Id auto increment, primary key
 - Name
 - Bno, a foreign key to Branch table at CPS 3740 database
- Insert 2 records
- Add a new column salary after name column
- Change the salary to meaningful values
- Delete a record

Exercises - view

- View features, advantages, disadvantages
- Create views
- Create non-updatable views
 - Why

Exercises

View and updatable view

- Create a view
 - Show all the female in the Staff at CPS3740 database
 - Is this view updatable, explain your reasons
- Create a view
 - Show the number of staff in each branch
 - Is this view updatable, explain your reasons

Exercise – outer join, subquery, SET operations

- Outer query
 - Left, right
- Subquery
- SET – Union, intersect, except

Relational Algebra Exercise

- Use the following table (DB **CPS3740** on imc)
- Please convert the following relational algebra operations to SQL and write its output:

T_Employee				
E_ID	Name	Dept	Salary	L_ID
1	Austin	IT	50000	1
2	Mary	HR	45000	3
3	Sam	IT	53000	2
4	Andrew	Sales	42000	1

$\rho_R(\text{myDept}) \mathfrak{I}_{\text{COUNT Dept}} (\text{T_Employee})$

Select count(Dept) as myDept
from T_Employee;

```
+-----+  
| myDept |  
+-----+  
|      4 |  
+-----+
```

More Exercises

- SQL exercise posted on the Google drive
- SELECT: 3, 17, 18, 19, 25
- Aggregate, GROUP BY: 26, 31, 40, 41, 42
- Subquery: 45, 48, 53, 57
- Joins: 58, 62, 67, 68, 71, 72
- SET: 73, 74, 76
- INSERT, UPDATE, DELETE: 77, 79, 81, 82, 83, 85
- CREATE, ALTER, DROP: 86, 87, 88, 103
- View and updatable view: 91, 110, 112, 113
- NULL operations: 116

Conclusions

- Study for midterm 2
- Must do all the exercises, NOT just read the slides
- Arrive classroom 5 minutes earlier